Day - 15
DICTIONARIES in Python

SKILL COURSE

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What is a Dictionary?

A dictionary is like a real dictionary book:

You search a **word (key)**

You get its **meaning (value)**

A Python dictionary works the same way:

Key = Word

Value = Meaning

So, dictionary stores data in **key–value pairs**.

→ Technical Definition

A dictionary is a collection of **unordered, mutable**, and **indexed key–value pairs** enclosed in { }.

→ Example:

{"name": "Satish", "age": 25, "city": "Mumbai"}

Why Dictionaries Are Important for Data Analysts?

- Store structured data
- Store column-wise data (like JSON)
- Map incorrect values to correct values
- Count occurrences (frequency table)
- Faster lookups
- Convert CSV rows into dictionary format
- Used in APIs, JSON files, database results
- Used in Pandas dictionaries, .to_dict(), JSON parsing

Creating a Dictionary & Accessing

```
student = {
    "name": "Rohit",
    "age": 21,
    "city": "Pune"
}

print(student["name"])
print(student["age"])
```

Accessing Values Using Keys

- **Adding New Key–Value Pairs**
  - student["marks"] = 85
- **Updating Existing Values**
  - student["city"] = "Mumbai"
- **Removing Items**
  - student.pop("age") # remove by key
  - student.clear() # remove all

Looping Through Dictionary

Loop keys

```
for k in student:
    print(k)
```

Loop values

```
for v in student.values():
 print(v)
```

Loop key–value pairs

```
for k, v in student.items():
    print(k, ":", v)
```

Nested Dictionaries (Real Use Case)

- employees = {
- "E101": {"name": "Rohit", "city": "Pune"},
- "E102": {"name": "Sneha", "city": "Delhi"}
- }

- **Updating Nested Dictionary**
  - employees["E101"]["city"] = "Mumbai"

Dictionary for Mapping Wrong → Correct Values

```
wrong_to_correct = {
    "mumbay": "Mumbai",
    "mombai": "Mumbai",
    "bengluru": "Bengaluru"
}

city = "mombai"

print(wrong_to_correct[city])
```

Dictionary for Counting Frequency (Real DA Work)

```
words = ["apple", "apple", "banana", "mango", "banana"]
freq = {}
for w in words:
    if w in freq:
        freq[w] += 1
    else:
        freq[w] = 1
print(freq)
```

Output:
{'apple': 2, 'banana': 2, 'mango': 1}

JSON-Like Dictionary (Used Everywhere)

```
product = {
    "id": "P102",
    "title": "Laptop",
    "price": 45000,
    "details": {
        "brand": "Dell",
        "color": "Black"
    }
}
```